

Spread <= \$0.04	(Proposed Reduction to \$0.005 or smaller)	4,333	84.8%	66.5%
Price < \$1:				
	\$0.0001	1,106	12.3%	0.1%
^a In this table, quoted spreads, and thus tick sizes, are determined by computing the time-weighted quoted spread during regular trading hours as computed by the WRDS intra-day indicators for every sym_root and sym_suffix combination in the WRDS intra-day indicators dataset and taking the equal weighted average across all trading days in January - March 2023. Stocks with average quoted spreads less than \$0.015 are assigned a \$0.005 tick. All other stocks are assigned a \$0.01 tick. A stock with a price less than \$1.00 will still be assigned a tick size per the usual process, which would be in force should the stock's price rise above \$1.00. As long as the stock's price remains below \$1.00 the \$0.0001 tick size would prevail. The designated tick size is applied to trading volume in in May – October 2023 where share and dollar volume is obtained from the universe of stocks in WRDS intra-day indicators. New stocks are given a tick size of \$0.01. The number of stocks assigned to each group is indicated in the <i>Number of Stocks</i> column and indicates the average number of stocks in each category (listings and delistings can affect the daily number of stocks trading as well as if a stock's price falls below \$1). If a stock has a VWAP of less than \$1.00, then that stock, as well as all of its trading volume for that day, is assigned to the \$0.0001 tick size. This estimate may be an upper bound. As discussed in section VII.B.3, <u>supra</u> and <u>infra</u> section VII.D.2, rebates can lower the quoted spread (although not necessarily transaction costs). Thus, lowering the access fee, and thus the associated rebates, may lead to wider quoted spreads. Because of this, some stocks may have quoted spreads that meet the threshold for the smaller tick size in the current environment but may not meet that threshold once the access fee cap is reduced, leading to lower rebates offered. Additionally, all stocks, even those priced below \$1.00, will be assigned a tick size via the usual process. If a stock price falls below \$1.00 the applicable tick size will be \$0.0001. So not all stocks initially assigned the \$0.005 tick size will trade differently than the baseline. This table differs from Table 3 because Table 3 is based on daily average TWAQs and does not attempt to analyze the effect of the adopted amendments. Once implemented, the changes to the current arrangements for consolidated market data pursuant to the MDI Rules may impact the number of stocks and their estimated percentage volumes anticipated for each tick level. In particular, under the MDI Rules, NMS stocks priced \$250 or more will receive reductions in round lot sizes which is anticipated to lower their quoted spreads; however, the effect on the reported numbers is likely small both because these stocks make up less than 4% of share volume and because they are unlikely to have quoted spreads less than \$0.015. Based on an analysis of data from May-October 2023, the average quoted spread of a stock priced between				

\$250 and \$1,000 was \$0.71, far greater from the \$0.015 that will trigger a smaller minimum increment. Similarly, for stocks priced between \$1,000 and \$10,000 the average quoted spread was \$3.85 and the only stock that had a value weighted average price greater than \$10,000 already has a round lot size of one share and had an average quoted spread of \$0.07.

Table 7 indicates that, had the amendments been in place in 2023, approximately 66% of share volume and 43% of dollar volume, associated with an estimated 1,788 individual stocks, would likely have been assigned the \$0.005 tick size. The adopted Rules represent a significant reduction in the scope of the Rule compared to the proposal. Table 7 provides estimates of the number of stocks and volume that would have been affected if the Commission had implemented the Rule with the tick size thresholds as proposed (the proposal would have lowered the tick size for all stocks with TWAQS less than \$0.04). The Commission estimates that there would have been 4,333 stocks receiving a smaller tick accounting for 84.8% (66.5%) of share (dollar) volume if all stocks with a TWAQS less than or equal to \$0.04 received a smaller tick size. Consequently, the number of stocks receiving a lower tick size is more than halved under the adopted amendments.

b. Effects on Market Quality

For the stocks that will receive the \$0.005 tick, the Commission expects market quality to improve. Smaller tick sizes present a market quality tradeoff between increasing penny and complexity concerns – which can harm market quality – and reducing pricing constraints – which can improve market quality by reducing pricing distortions leading to an oversupply of liquidity relative to competitive levels. The Commission believes that market quality will, on average, improve for stocks receiving the smaller tick based on theoretical discussion, the Commission’s empirical analysis, as well as evidence and opinions expressed by commenters.

For example, one commenter agreed with the presence of market distortions under current tick sizes, stating: “[t]he SEC correctly describes the problem of tick-constrained securities. Such securities are ‘not able to be priced by market forces’ because the current ‘rule 612 minimum pricing increment of \$0.01 may now be too large for certain stocks, which, in turn, results in the pricing of such stocks being artificially constrained.’ Trading in these securities would be improved ‘if competitive market forces could establish prices in sub-penny increments, which could reduce quoted spreads,’ allowing these securities to ‘be priced more aggressively within the spread.’”¹¹⁹⁵

The theoretical discussion provided below supports characterizing a smaller tick size as providing a penny/complexity versus pricing constraint tradeoff, and the empirical analysis presented in Table 8 as well as other empirical research suggests that, for stocks with fewer than approximately two ticks intra-spread,¹¹⁹⁶ a reduction in the tick size on average improves market quality. A number of commenters agreed, and some commenters presented analyses suggesting that 2 to 4 ticks intra-spread may be optimal. Combined, this evidence suggests that the tick size reduction associated with these amendments will, on average, improve market quality for the subset of stocks receiving the lower tick size.¹¹⁹⁷

¹¹⁹⁵ See Nasdaq Letter I at 11 (quoting the Proposing Release).

¹¹⁹⁶ We use the terminology “ticks intra-spread” or “ticks within the spread” to mean the number of quoting increments between the NBB and NBO (the quoted spread). For example, if the quoted spread is one penny wide (in a stock priced above \$1), then we say that there is one tick intra-spread under the baseline. Under the baseline, symbols priced above \$1.00 with a quoted spread between 2 and 4 pennies would have 2 to 4 ticks intra-spread.

¹¹⁹⁷ The amendments will take stocks trading with 1-1.5 ticks intra-spread and increase the number of ticks intra-spread to up to 3.

i. Theoretical Discussion

Tick sizes present an economic tradeoff.¹¹⁹⁸ All else equal, reducing the tick size improves market quality by reducing distortions associated with markets not being able to set prices that equate liquidity supply and demand in the presence of a discrete pricing grid.¹¹⁹⁹ In a competitive market, and in the absence of rebates or other price distortions, the prevailing bid or ask price will be the feasible price equal to or just worse than the price that equates supply and demand for the underlying asset.¹²⁰⁰ This is because liquidity providers will not post bids and offers that would result in guaranteed trading losses – *i.e.*, they will not post prices that do not bring in sufficient revenue to cover their marginal cost of providing liquidity.¹²⁰¹ Since there is competition along a finite pricing grid, they choose the closest feasible price just worse than the competitive one. The gap between the feasible price and the price that equates liquidity supply and demand – *i.e.*, the competitive price – is a price distortion allowing liquidity providers to earn rents on liquidity provision.

This pricing distortion is most relevant for stocks that are tick-constrained and diminishes as quoted spreads widen. To understand this, consider again the example of section VII.B.2. In that example, under a tick size of \$0.005, the ask would be \$10.015 and the bid \$10.005. However, with a tick size of \$0.01, the ask would be \$10.02 and the bid \$10.00, implying a

¹¹⁹⁸ See section VII.B.2

¹¹⁹⁹ See, *e.g.*, Rindi and Graziani Letter at 2 (agreeing), see also Barardehi et al., *supra* note 231 (for a more thorough discussion of this tradeoff). See also NASAA Letter at 9 (stating that the general concept that a narrower tick size will increase pricing efficiency), as well as discussion in Ingrid M. Werner, et al., *Tick Size, Trading Strategies and Market Quality*, 69 MGMT. SCI. 3818 (2023). See also Budish Letter at 4 referring to a tick size that is too wide as producing rents via regulatory price constraints.

¹²⁰⁰ Any price better than this will lead to an excess of liquidity demand which will push prices out again.

¹²⁰¹ Marginal cost in this context refers to the cost of providing an additional share of liquidity. If the revenue associated with providing a share of liquidity is less than the cost of providing that share, then liquidity providers are better off not providing liquidity than incurring a loss to provide liquidity.

spread that is twice as wide. Now assume that the same issuer reduced the number of shares so that the stock increases in price 100-fold, but the underlying economics are the same. To achieve the same reduction in spread would not require any change to the tick size: an ask of \$1,001.50 and a bid of \$1,000.50 are feasible even with a tick size of one penny.

While a smaller tick size increases competition, thereby reducing distortions and reducing transaction costs, there are potential costs raised in the proposing release and also by commenters which are discussed below.

Pennying: The proposing release and commenters identified pennyning as a risk of a smaller tick which can harm market quality.¹²⁰² Pennyning occurs when limit order providers get to the front of the queue by providing economically trivial price improvement. It reduces the importance of time priority.¹²⁰³ The risk of being pennied could discourage liquidity provision in lit markets, particularly by market participants that are slower to respond to changes in market conditions and could increase transaction costs for these investors.¹²⁰⁴ To compensate for additional costs associated with a fragmented order book, liquidity providers may post less aggressive quotes leading to wider quoted spreads and worse market quality.¹²⁰⁵

Market participants may respond to an increased risk of pennyning by increasing their use of hidden or off-exchange orders that do not display prices, and thus avoid exposing the price

¹²⁰² See, e.g., Robinhood Letter at 41, Virtu Letter II at 4, Tastytrade Letter at 20, AIMA Letter at 2, Brandes Letter at 2, UBS Letter at 10, and TradeStation Letter at 5, Lewis Letter attached to Virtu Letter II at p 33. See also supra note 994 and section VII.B.2 for a definition and discussion of pennyning. See also Proposing Release, supra note 11 at section V.D.1.

¹²⁰³ See, e.g., Antitrust Division of the DOJ Letter at 5 and XTX Markets Letter at 3.

¹²⁰⁴ See Dyhrberg et al., supra note 994 studying the effects of imposing a tick size on a crypto exchange that previously did not have a tick size. The authors report an improvement in market quality due largely to a reduction in pennyning behavior. See also Virtu Letter II at 25 and Better Markets Letter I at 8. See also Budish Letter at 5.

¹²⁰⁵ See, e.g., Virtu Letter II at 8, Fidelity Letter at 10.

needed to beat in order to get to the front of the queue and increase the likelihood of a fill.¹²⁰⁶

Increased use of hidden orders has been associated with worse market quality outcomes.¹²⁰⁷

Some commenters expressed their belief that a narrower tick and increased pennyning could lead some orders that previously were at protected prices to be traded through.¹²⁰⁸ However, it is not clear from the commenters' letters why this would occur given the order protection rule and broker's best execution responsibilities. One commenter also suggested that narrow ticks could increase volatility.¹²⁰⁹ However, existing research on the topic would suggest, if anything, an opposite effect.¹²¹⁰

In contrast to the tick size pricing distortion discussed above, which is most relevant for stocks that are tick-constrained,¹²¹¹ the pennyning effect will be most pronounced for stocks with wide quoted spreads because there are more intra-spread price levels and the cost of gaining priority over other liquidity providers, by updating the best price by a single tick, is lower with a smaller tick.¹²¹² For example, a stock with a quoted spread of ten cents, and a \$0.01 tick, will have 10 price levels within the quoted spread, whereas a stock with a \$1.00 quoted spread and a

¹²⁰⁶ See, e.g., IEX Letter I at 12, Danny Mulson Letter at 1, Nasdaq Letter I at 2.

¹²⁰⁷ See, e.g., Amy K. Edwards, et al., The Effect of Hidden Liquidity: Evidence from an Exogenous Shock (working paper Mar. 1, 2021), available at <https://ssrn.com/abstract=3766512> (retrieved from SSRN Elsevier database) ("Edwards, et al. (2021)"). See also Danny Mulson Letter at 3 stating that a smaller tick size would lead to more hidden orders, specifically 'peg offset dark orders' which could harm price efficiency.

¹²⁰⁸ See Themis Letter at 5, Virtu Letter II at 6, 10 discussing how a smaller tick can weaken protected quotes.

¹²⁰⁹ See, e.g., Virtu Letter II at 8. See, also Edwin Hu et al., 2018; supra note 1002; and Kee H. Chung et al., Tick Size Liquidity for Small and Large Orders and Price Informativeness: Evidence From the Tick Size Pilot Program, 136 J. FIN. ECON. 879 (2020), who both report the opposite effect in the context of the Tick Size Pilot where stocks with wider ticks experienced more volatility.

¹²¹⁰ See id. see also e.g., Edwards, et al., (2021), supra note 1207.

¹²¹¹ See supra this section.

¹²¹² The pennyning effect would be particularly acute for wide-quoted spread stocks with lower stock prices because a lower stock price reduces the amount of capital needed to supply a round-lot quote and hence make pennyning less capital intensive.

\$0.01 tick will have 100. Because price has first priority in order execution, in a price-time priority system where quote priority is awarded based on best price first and then arrival order second, a primary way to gain priority for a trader providing liquidity is to price-improve over existing orders. Without a small tick size relative to the quoted spread, getting to the front of the queue via price improvement will be more costly, requiring larger relative price concessions.¹²¹³ Because the (beneficial) pricing efficiency effect is greatest when quoted spreads are narrow, whereas the (detrimental) pennyng effect is greatest when quoted spreads are wide, this analysis suggests setting a minimum quoting increment on the basis of average spread. Commenters agreed.¹²¹⁴

Fragmenting liquidity: The proposing release and commenters also discussed a cost of a lower tick size as spreading the displayed orders over more price levels.¹²¹⁵ When tick increments are farther apart, all else equal, liquidity providers that may have various prices at which they are willing to provide liquidity must congregate their quotes at only the available quoting increments. Thus, there will be more depth at each level including at the NBBO. With more price levels due to a smaller tick size, market participants can more accurately tailor their quotes to the prices at which they are willing to provide liquidity and thus liquidity will naturally

¹²¹³ For example, if a stock has a quoted spread of ten cents and a \$0.01 tick, gaining priority through price improvement would require narrowing the half-quoted spread (*i.e.*, the distance between the current quote and the midpoint) by 20%. If instead a stock has a quoted spread of \$1.00 with a \$0.01 tick, a market participant would only need to improve the half-quoted spread by 2% to get to the front of the queue.

¹²¹⁴ See Budish Letter at 4 and Harris Letter at 7 supporting the use of quoted spread as the determinate of the tick size. See also *infra* section VII.D.1.b.iii.

¹²¹⁵ See Proposing Release, *supra* note 11 at section V.D.1 for a discussion of fragmenting liquidity. See also *e.g.*, GTS Letter at 5 and CCMR Letter at 24.

spread over more levels and there will be fewer resting orders at each price level, including the NBBO.¹²¹⁶

Fragmenting liquidity across multiple price levels may decrease costs associated with smaller orders, which would be able to source liquidity at improved prices due to a finer price grid.¹²¹⁷ However, it can increase the complexity and cost associated with sourcing liquidity for larger orders,¹²¹⁸ as the reduction in shares available at the top of the book will render it more likely that a market participant must source liquidity beyond the NBBO in order to execute a trade.¹²¹⁹ It could also increase the number of child orders a parent order needs to be divided into to execute, which could increase the overall complexity and likelihood of information leakage leading to increased transaction costs via increased price impact.¹²²⁰

Quote Instability: Commenters also stated that less depth at the NBBO can lead to increased NBBO quote instability as trades are more likely to deplete depth at the NBBO prices.¹²²¹ One commenter presented evidence that lower quote stability is empirically associated

¹²¹⁶ See, e.g., GTS Letter at 5, CCMR Letter at 24, and UBS Letter at 11.

¹²¹⁷ See UBS Letter at 11.

¹²¹⁸ Id. See also TradeStation Letter at 6 mentioning as an example of increased complexity that brokers would have to put systems in place to manage customers' good-till-canceled trades that may remain open over a weekend when a tick size change is implemented. See also discussion in Lewis Letter attached to Virtu Letter II at 34-35.

¹²¹⁹ See, e.g., IEX Letter I at 13 discussing how order shredding with smaller ticks can increase information leakage, such as when quotes on other exchanges are cancelled when limit orders on one exchange begin to be executed, potentially signaling a large price moving trade. Brandes Letter at 2 states that increased complexity associated with more pricing increments would be to the detriment of longer-term investors. Equity Market Structure Citadel Letter at 2 states that for institutional investors, "[l]arger orders will be more complex to execute, as filling the entire order will require accessing multiple price levels, which can increase price impact."

¹²²⁰ See, e.g., Citadel Letter I at 5, 9 and Virtu Letter II at 8, 10 discussing the price impact of large trades under a regime of smaller ticks; stating that smaller ticks could increase price impact.

¹²²¹ Some commenters stated that smaller ticks would lead to more "flickering quotes," which are defined in the Reg NMS release as quotes that flashed for a short period of time solely to earn market data revenues, but were not truly accessible and therefore did not add any value to the consolidated quote stream. However,

with increased market making costs, which it states may deter liquidity provision.¹²²² While increased quote instability may occur in stocks receiving the lower tick, the lower tick itself will allow market forces to adjust the price of liquidity – i.e., the quoted spread – such that market makers are competitively compensated for the risks associated with providing liquidity. Increased instability in the NBBO could make it more difficult to determine which exchange has the best price at a given point in time and thus where to route an order.¹²²³ This could be particularly true when markets are volatile.¹²²⁴

Increasing (or decreasing) rents to speed: The Proposing Release stated that “too small ticks may inefficiently award speed”.¹²²⁵ As discussed in the next few paragraphs, commenters also commented on the effects of tick size on speed. Investments in speed are a fixed and largely irreversible cost that some market participants choose to incur. Changing the tick size could change the profitability of such investments, that is, they could increase or decrease the rents to speed. As noted in section VIII.B.2, a narrower tick reduces rents that accrue when a liquidity provider can be first in line in a queue.¹²²⁶ That is, narrowing the tick would be expected to

the Commission believes that this is unlikely for reasons discussed in the Proposing Release note 195 and surrounding text relating to advances in exchange technology. Other commenters defined ‘flickering quotes’ more broadly simply as periods of time where the NBBO changes rapidly, see, e.g., IEX Letter I at 8 and Robinhood Letter at 20. Much of the concern these commenters expressed was with respect to the proposed \$0.001 and \$0.002 tick sizes which are not part of the adopted amendments, see IEX Letter I at 8 and Robinhood Letter at 20. As discussed here, the Commission acknowledges that a smaller tick will likely lead to more frequent changes to the NBBO and discusses those consequences herein.

¹²²² See IEX Letter I at 11-12.

¹²²³ See, e.g., Themis Letter at 6 and CCMR Letter at 17 and MFA Letter at 1. Quote instability could increase the complexity associated with complying with Rule 611 as it could make it harder to determine which exchange currently has the best price.

¹²²⁴ See Citadel Letter I at 2, 7.

¹²²⁵ See Proposing Release, supra note 11, at 80306.

¹²²⁶ See, e.g., Budish Letter at 1 (“Reducing the tick-size constraint for tick-constrained stocks will reduce excess rents from artificially constrained prices. These excess rents lead to a speed race to the top of the book, which increases complexity, and the rents come at the expense of investors via a higher cost of liquidity.”), Antitrust Division of the DOJ Letter at 5, and XTX Markets Letter at 3.

reduce rents to speed. However, speed confers an advantage in implementing a pennyning strategy: a trader can not only step ahead of another trader, but also potentially sell (or buy) an asset back to the other trader if the market moves unfavorably, replicating an option-like payoff.¹²²⁷ The amendments are limited to stocks with spreads for which pennyning is unlikely to be a dominant effect. Nonetheless, to the extent that pennyning increases, it has the potential to increase the rents to speed.

In the context of the proposal, one commenter stated that a smaller tick size would be expected to increase the frequency of sniping because smaller ticks generate faster and more frequent price changes.¹²²⁸ While there will be more prices at which to trade, the underlying information is not changing (prices may change more rapidly, but the information of each price change is smaller). That is, sniping may become more frequent, but the profits per each individual snipe attempt would decline. However, and as stated above, the Commission does agree that a large number of ticks within the spread can make pennyning more prevalent, and to the extent that profits are linked to speed, can increase the rents to speed. The adopted amendments imply fewer ticks intra-spread than the proposing amendments, reducing this effect. Thus, the Commission does not expect slower traders to be disadvantaged by the adopted amendments.

Effect on thinly traded securities: One commenter stated that narrower quoted spreads due to a smaller tick would be harmful for liquidity, particularly for smaller and medium-sized

¹²²⁷ See supra note 993 for a discussion on the relationship between pennyning and trading speed. See supra note 1202 and accompanying text for discussions regarding the amendments to Rule 612 and pennyning.

¹²²⁸ See Virtu Letter II at 23. Sniping pertains to the ability to “pick off”, by executing against a stale quote in response to new information before it can be updated.